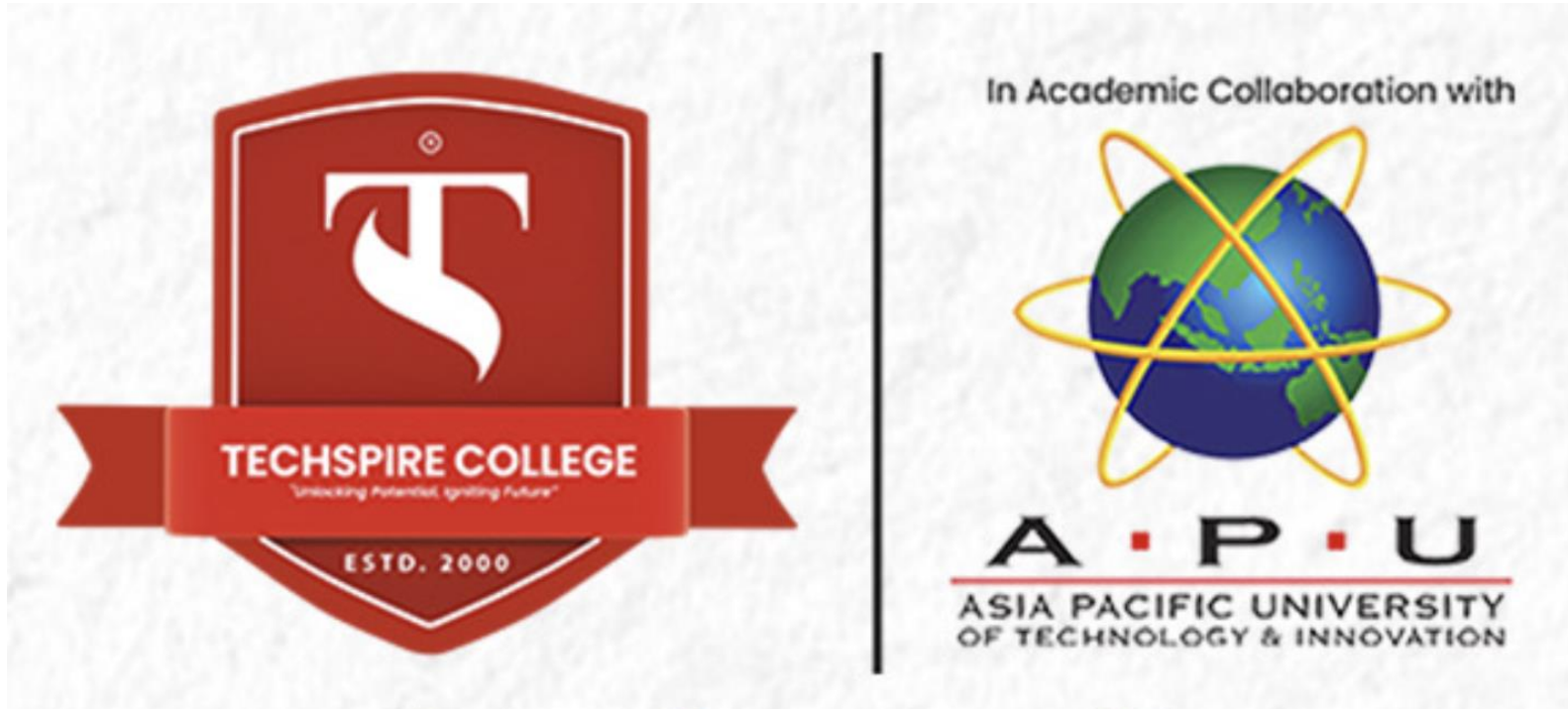




OOP in Java

Slide 3 – Decisions

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Making Decisions with Java: if Statements

- What Is a Decision in a Program?
 - Imagine you're choosing what to wear based on the weather. If it's sunny, you wear shorts; if it's raining, you wear a raincoat. In programming, decisions help the computer choose what to do based on certain conditions.
- Introducing the if Statement
 - The if statement allows the computer to make a choice. It's like asking a question and deciding what to do based on the answer.
 - **Condition:** This is a question or test that will be either **true** or **false**.

```
if (condition) {  
    // Code to run if the condition is true  
}
```

Example

- **Condition:** age $\geq$ 18
 - This checks if the age is 18 or more.
- **Action:** System.out.println("You are eligible to vote.");
 - This message is printed if the condition is true.

Example: Check if someone is old enough to vote.

```
int age = 18;  
if (age  $\geq$  18) {  
    System.out.println("You are eligible to vote.");  
}
```

Adding an else Option

- Sometimes, you want to do something different if the condition is not true. You use else for this.
- If age is less than 18, the message "You are not eligible to vote yet." will be printed.

Syntax

```
if (condition) {  
    // Code to run if the condition is true  
} else {  
    // Code to run if the condition is false  
}
```

Example: Check if someone is old enough to vote and provide feedback.

```
int age = 16;  
  
if (age >= 18) {  
    System.out.println("You are eligible to vote.");  
} else {  
    System.out.println("You are not eligible to vote yet.");  
}
```

If you have more than two choices, you can use else if to test additional conditions.

Adding More Choices with else if

Example: Check the age and print different messages based on the age range.

```
if (age < 13) {  
    System.out.println("You are a child.");  
} else if (age < 20) {  
    System.out.println("You are a teenager.");  
} else {  
    System.out.println("You are an adult.");  
}
```

```
if (condition1) {  
    // Code to run if condition1 is true  
} else if (condition2) {  
    // Code to run if condition2 is true  
} else {  
    // Code to run if none of the above conditions are true  
}
```

Exercise

A thick yellow horizontal bar spans the width of the slide, with a vertical yellow bar extending downwards from its right end.

- **Task 1:** Write a program to check if a number is positive, negative, or zero.
- **Task 2:** Create a program to determine if a person is eligible for a senior citizen discount (60 years or older).



Summary

- **if Statement:** Allows the program to make decisions based on conditions.
 - **else:** Provides an alternative action if the condition is false.
 - **else if:** Adds more choices for multiple conditions.
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